BITS Pilani

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CE F231
Fluid Mechanics

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Lecture 34: Flow Through Pipes

Recap: Major losses

- Losses due to friction

  - Hagen-Poiseuille equation → viscous flow

$$h_f = \frac{32\mu\bar{u}L}{\rho g D^2}$$

  - Darcy-Weisbach equation → viscous and turbulent flow

$$h_f = \frac{4\,.\,f'}{2g}\cdot\frac{LV^2}{d}$$

or

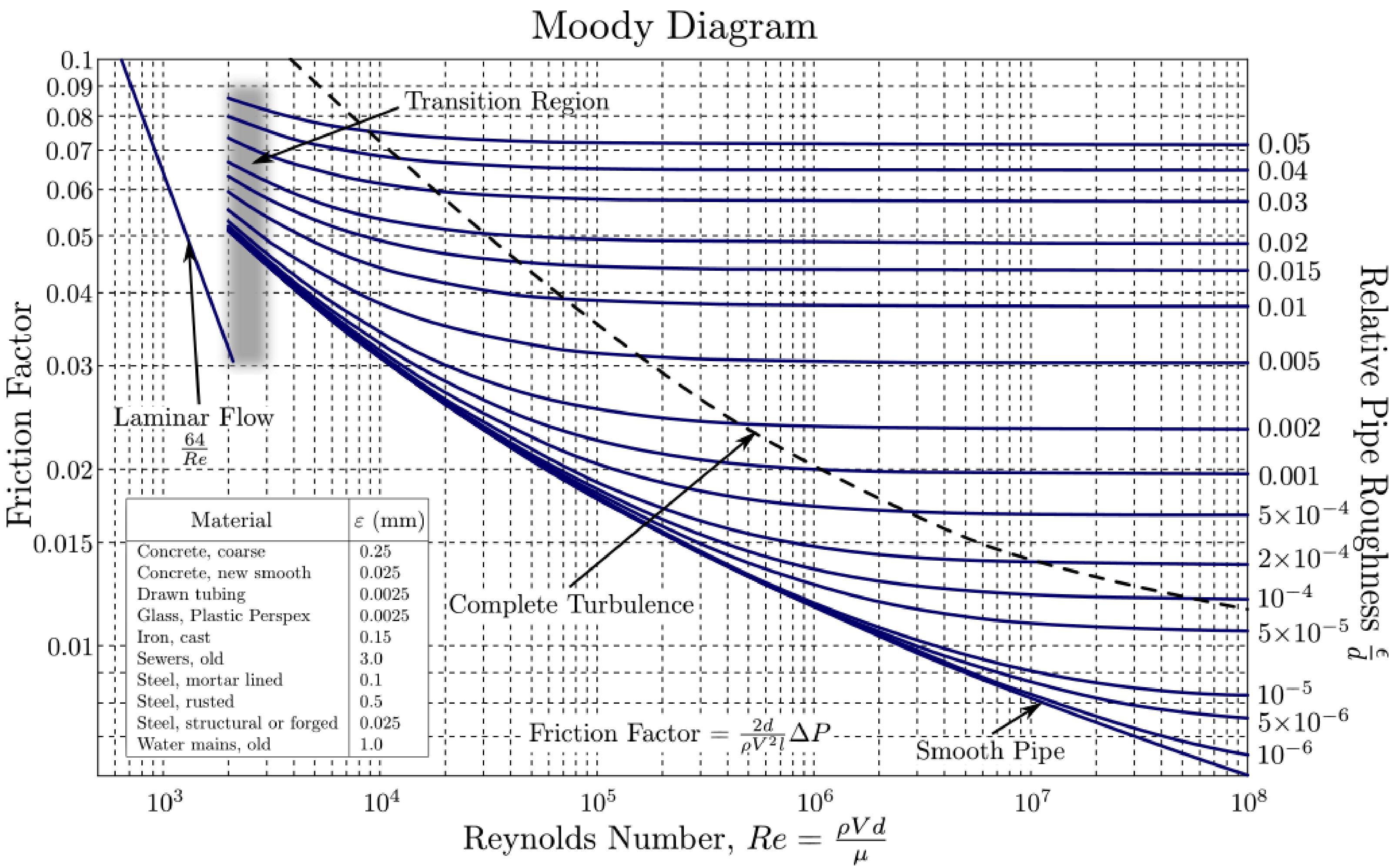
$$h_f = \frac{f\,.\,L\,.\,V^2}{d \times 2g}$$

f' = coefficient of friction or Fanning friction factor

f = friction factor

Recap: Major losses

- Losses due to friction

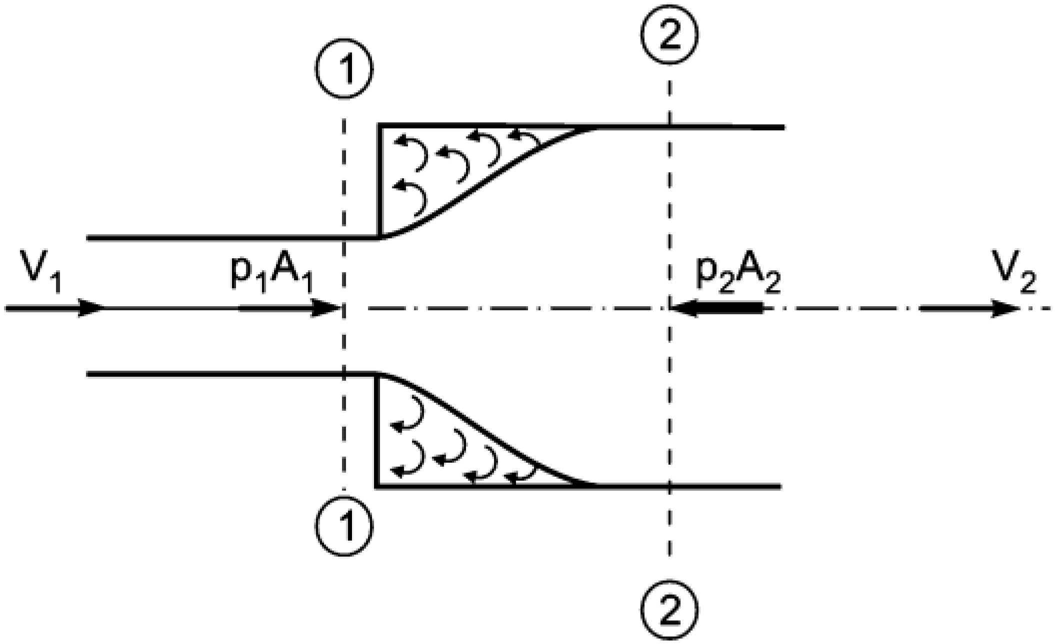


Material	ε (mm)
Concrete, coarse	0.25
Concrete, new smooth	0.025
Drawn tubing	0.0025
Glass, Plastic Perspex	0.0025
Iron, cast	0.15
Sewers, old	3.0
Steel, mortar lined	0.1
Steel, rusted	0.5
Steel, structural or forged	0.025
Water mains, old	1.0

https://en.wikipedia.org/wiki/Moody_chart

Recap: Minor losses

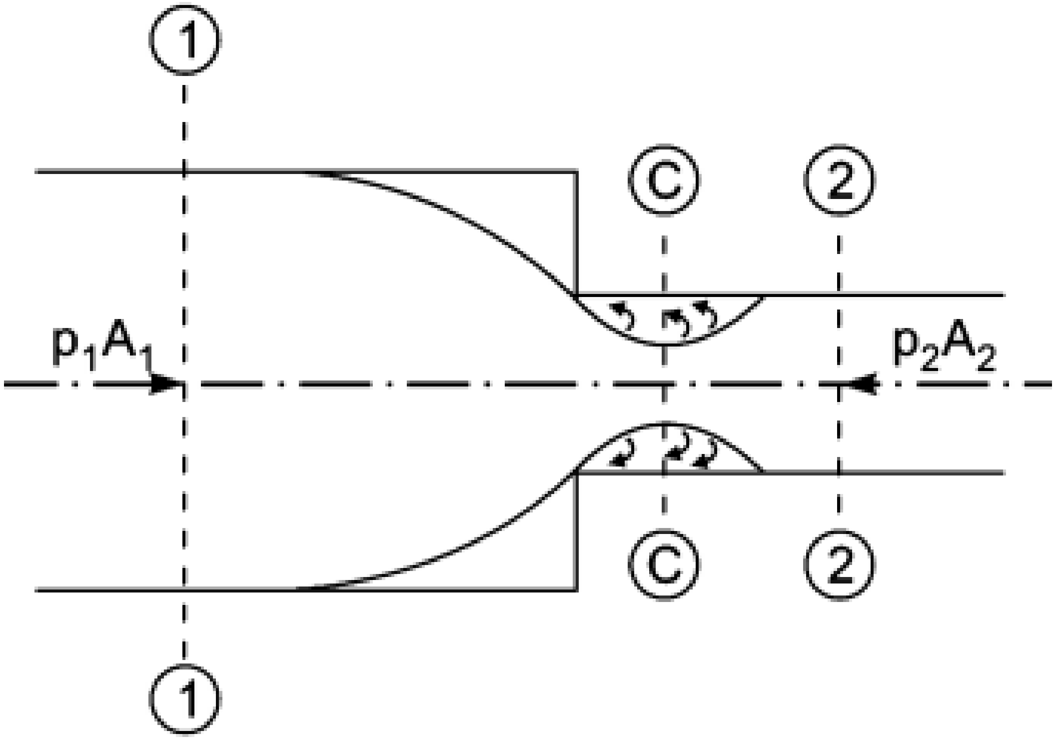
- Losses due to sudden enlargement



$$h_e = \frac{(V_1 - V_2)^2}{2g}$$

Recap: Minor losses

- Losses due to sudden contraction



$$h_c = \frac{V_2^2}{2g}\left[\frac{1}{C_c} - 1\right]^2$$

$$h_c = 0.5\,\frac{V_2^2}{2g}$$

(Used in general)

Recap: Minor losses

- Head loss at entrance of pipe

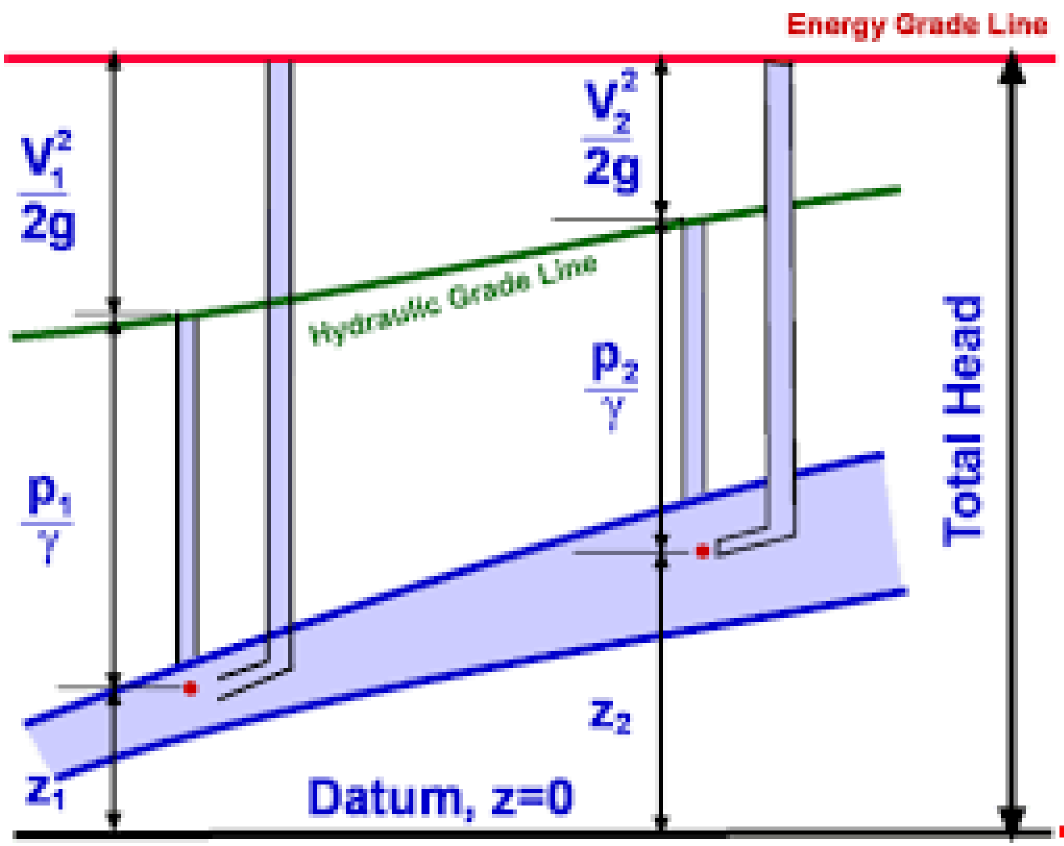
$$h_i = 0.5\,\frac{V^2}{2g}$$ (≡Sudden contraction)

- Head loss at exit of pipe

$$h_o = \frac{V^2}{2g}$$ (≡Sudden enlargement)

Hydraulic grade line and energy grade line

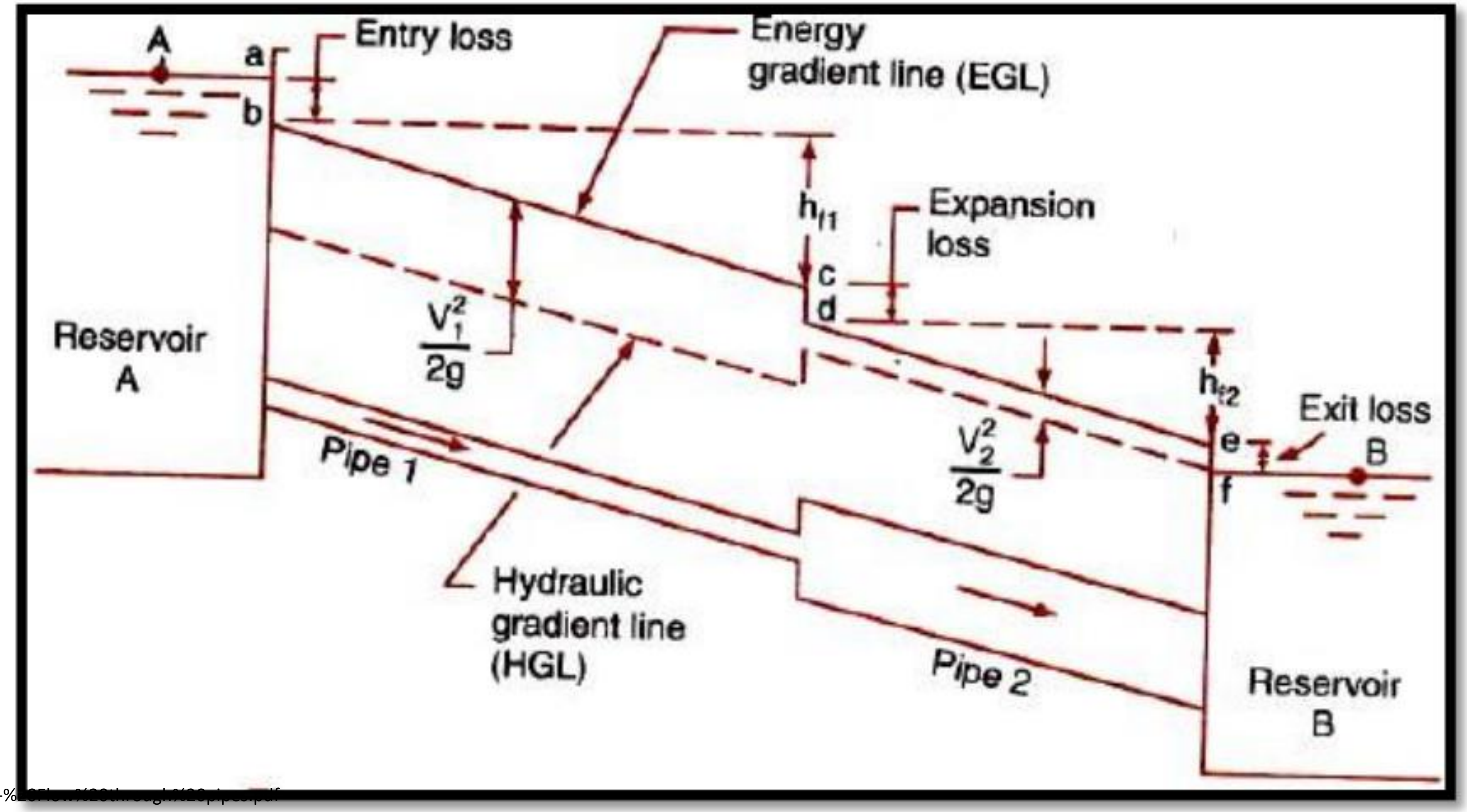
- Hydraulic grade line – joining pressure + datum heads
- Energy grade line – joining pressure + datum + kinetic heads



http://www-mdp.eng.cam.ac.uk/web/library/enginfo/aerothermal_dvd_only/aero/fprops/cvanalysis/node40.html

Hydraulic grade line and energy grade line

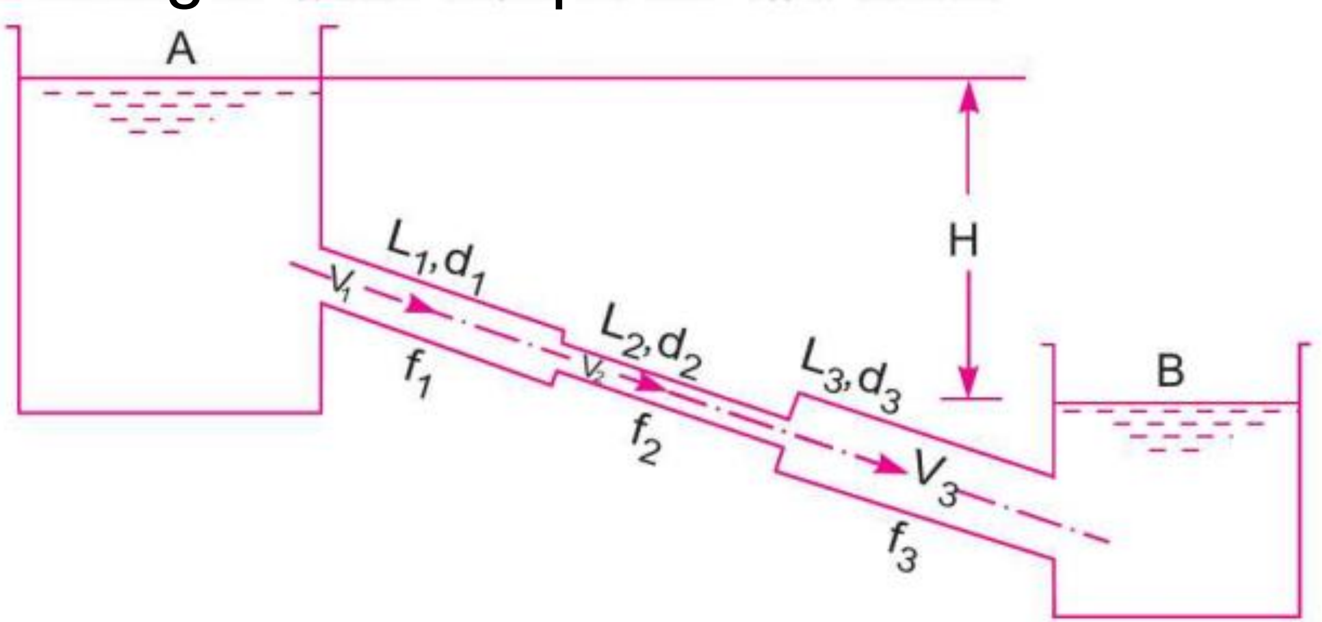
- Hydraulic grade line – joining pressure + datum heads
- Energy grade line – joining pressure + datum + kinetic heads



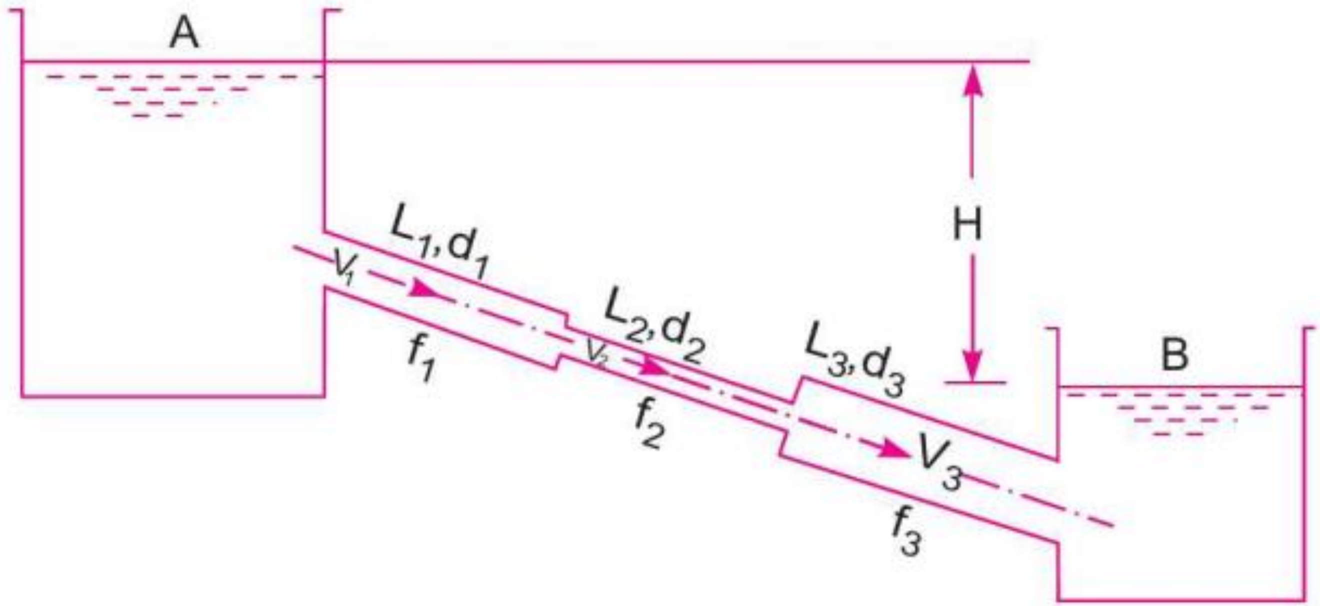
http://www.bsiet.org/notes/C3-%20Flow%20through%20pipes.pdf

Flow through pipes in series

- Pipes of different lengths and diameters are connected end-to-end
- Discharge passing through each pipe is same
- Head losses get added up



Flow through pipes in series



Continuity: conservation of mass → $Q = A_1V_1 = A_2V_2 = A_3V_3$

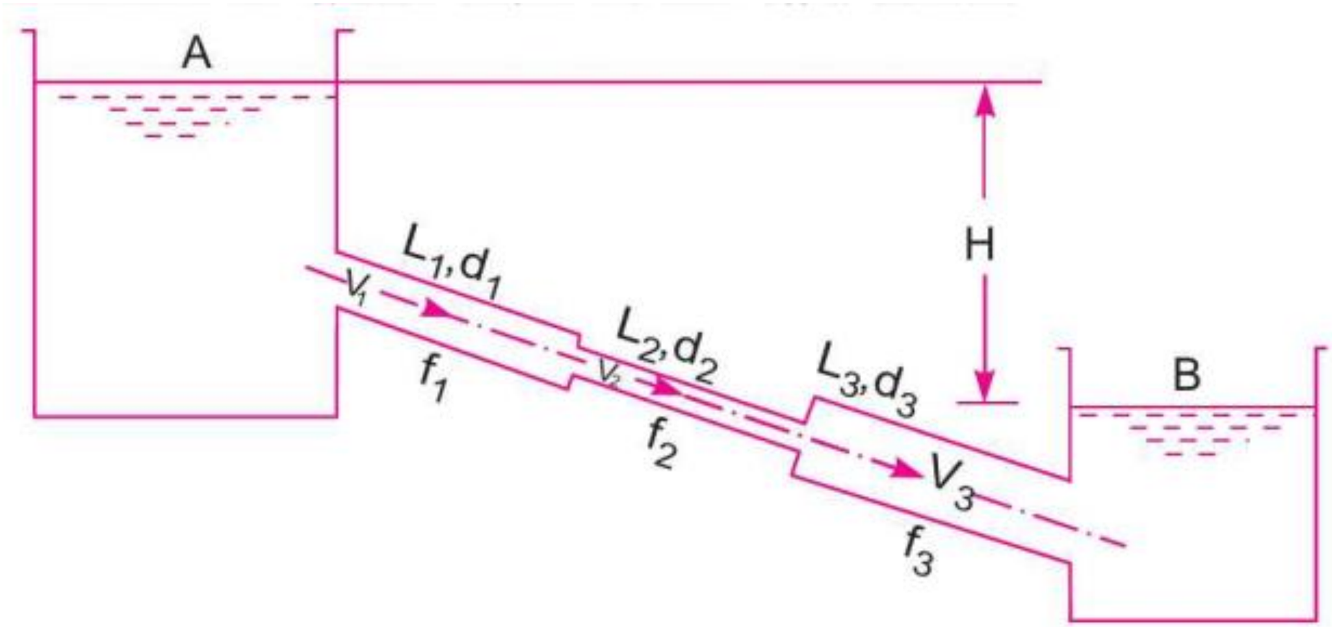
Total head loss = elevation difference in water surfaces in two reservoirs →

$$H = \frac{0.5\,V_1^2}{2g} + \frac{4 f_1 L_1 V_1^2}{d_1 \times 2g} + \frac{0.5\,V_2^2}{2g} + \frac{4 f_2 L_2 V_2^2}{d_2 \times 2g} + \frac{(V_2 - V_3)^2}{2g} + \frac{4 f_3 L_3 V_3^2}{d_3 \times 2g} + \frac{V_3^2}{2g}$$

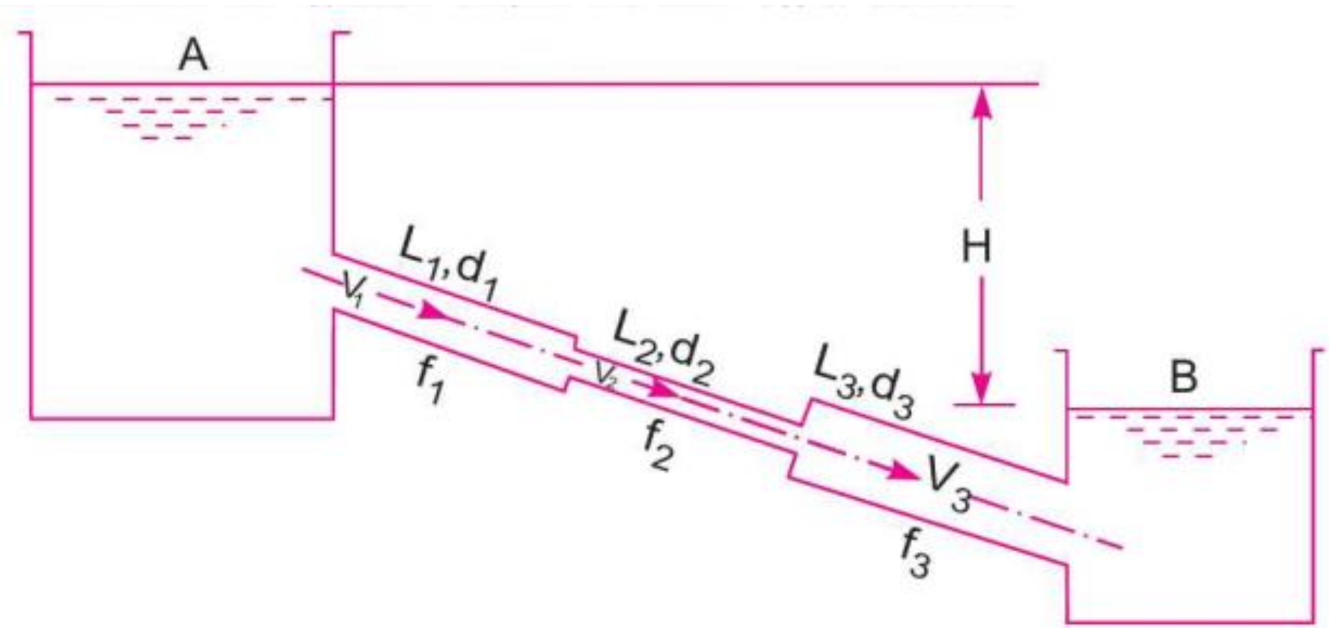
Equivalent pipe (for pipes in series)

➢Equivalent pipe of uniform diameter

- Head loss same as compound pipe (pipes in series)
- Discharge same as compound pipe (pipes in series)
- Length of equivalent pipe = total length of compound pipe



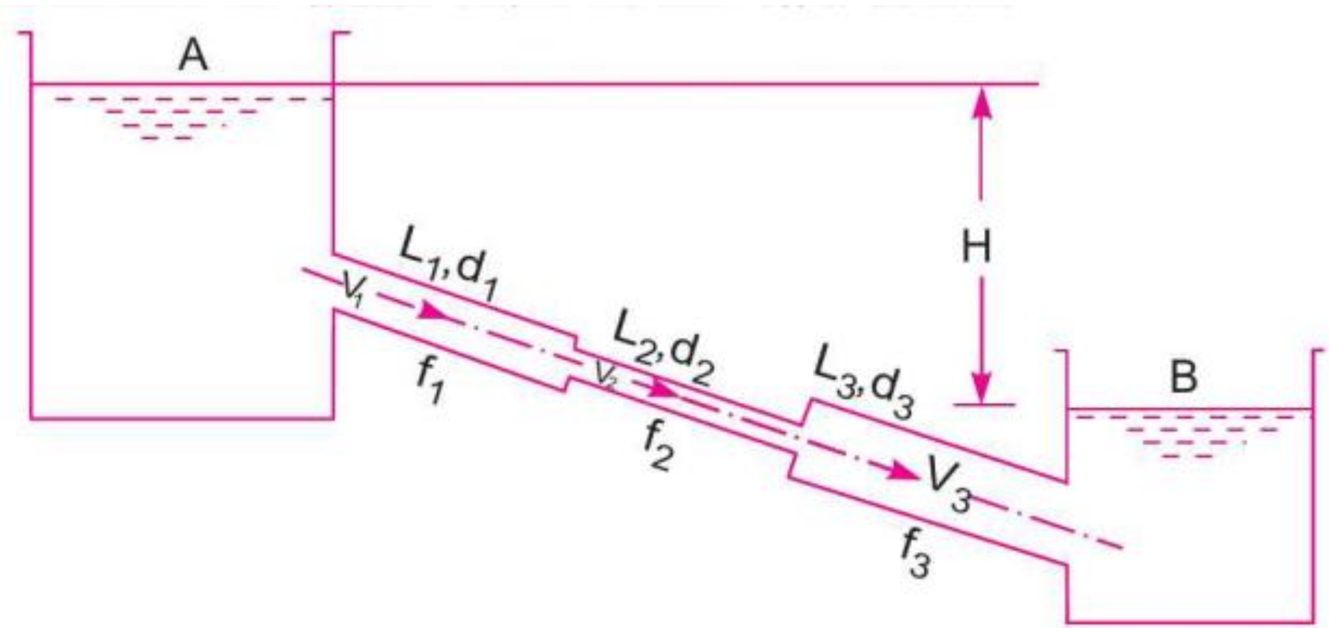
Equivalent pipe (for pipes in series)



Total head loss in the compound pipe, neglecting minor losses

$$H = \frac{4 f_1 L_1 V_1^2}{d_1 \times 2g} + \frac{4 f_2 L_2 V_2^2}{d_2 \times 2g} + \frac{4 f_3 L_3 V_3^2}{d_3 \times 2g}$$

Equivalent pipe (for pipes in series)

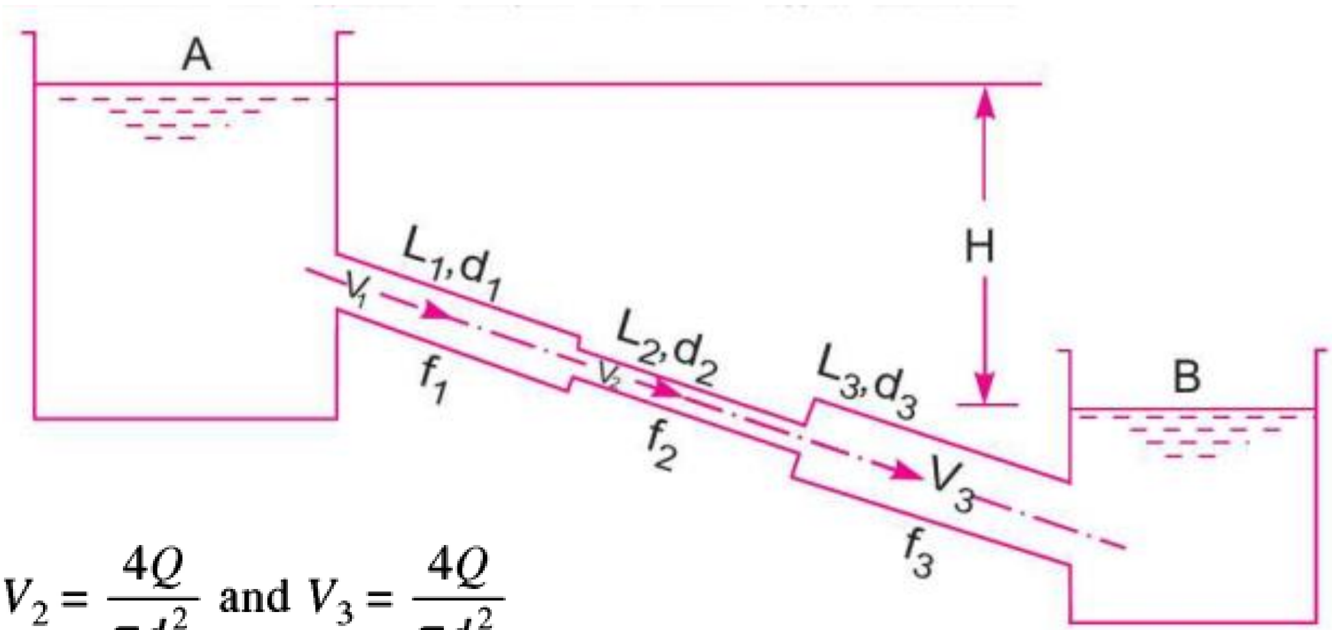


Assuming $$f_1 = f_2 = f_3 = f$$

Discharge, $$Q = A_1 V_1 = A_2 V_2 = A_3 V_3 = \frac{\pi}{4} d_1^{\,2} V_1 = \frac{\pi}{4} d_2^{\,2} V_2 = \frac{\pi}{4} d_3^{\,2} V_3$$

$\therefore$ $$V_1 = \frac{4Q}{\pi d_1^2},\ V_2 = \frac{4Q}{\pi d_2^2} \text{ and } V_3 = \frac{4Q}{\pi d_3^2}$$

Equivalent pipe (for pipes in series)

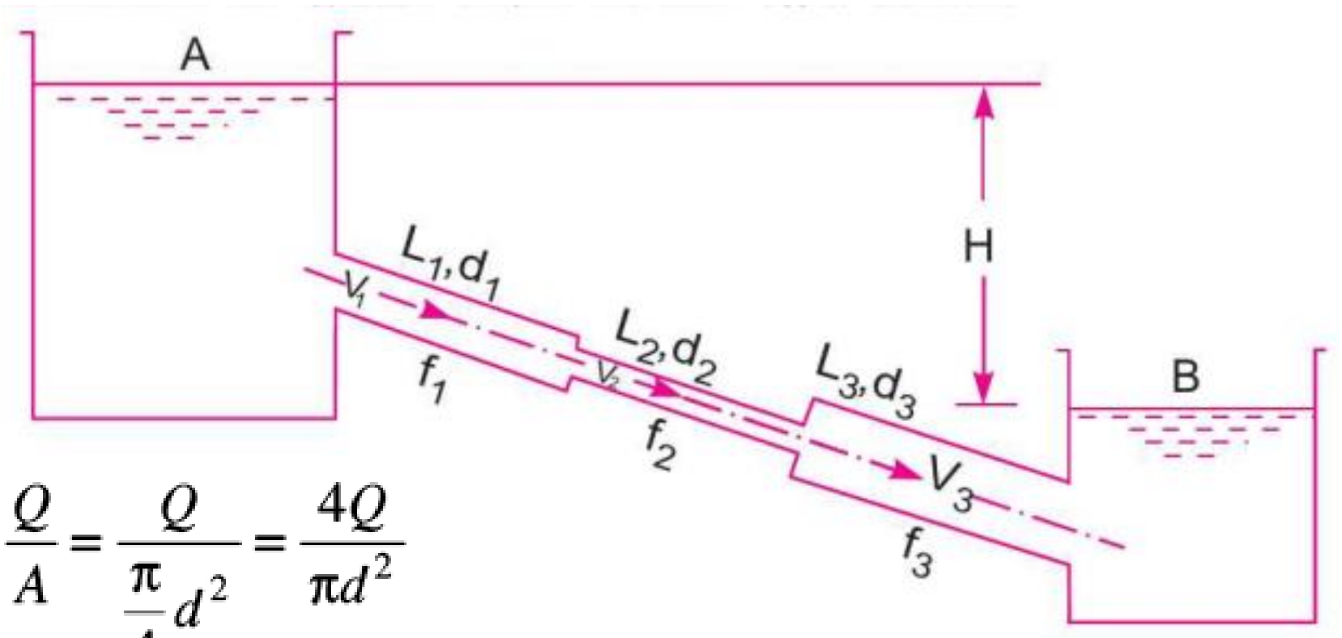


$$V_1 = \frac{4Q}{\pi d_1^2},\ V_2 = \frac{4Q}{\pi d_2^2} \text{ and } V_3 = \frac{4Q}{\pi d_3^2}$$

$$H = \frac{4 f_1 L_1 V_1^2}{d_1 \times 2g} + \frac{4 f_2 L_2 V_2^2}{d_2 \times 2g} + \frac{4 f_3 L_3 V_3^2}{d_3 \times 2g} \longrightarrow H = \frac{4 f L_1 \times \left(\frac{4Q}{\pi d_1^2}\right)^2}{d_1 \times 2g} + \frac{4 f L_2 \left(\frac{4Q}{\pi d_2^2}\right)^2}{d_2 \times 2g} + \frac{4 f L_3 \left(\frac{4Q}{\pi d_3^2}\right)^2}{d_3 \times 2g}$$

$$= \frac{4 \times 16 f Q^2}{\pi^2 \times 2g}\left[\frac{L_1}{d_1^5} + \frac{L_2}{d_2^5} + \frac{L_3}{d_3^5}\right]$$

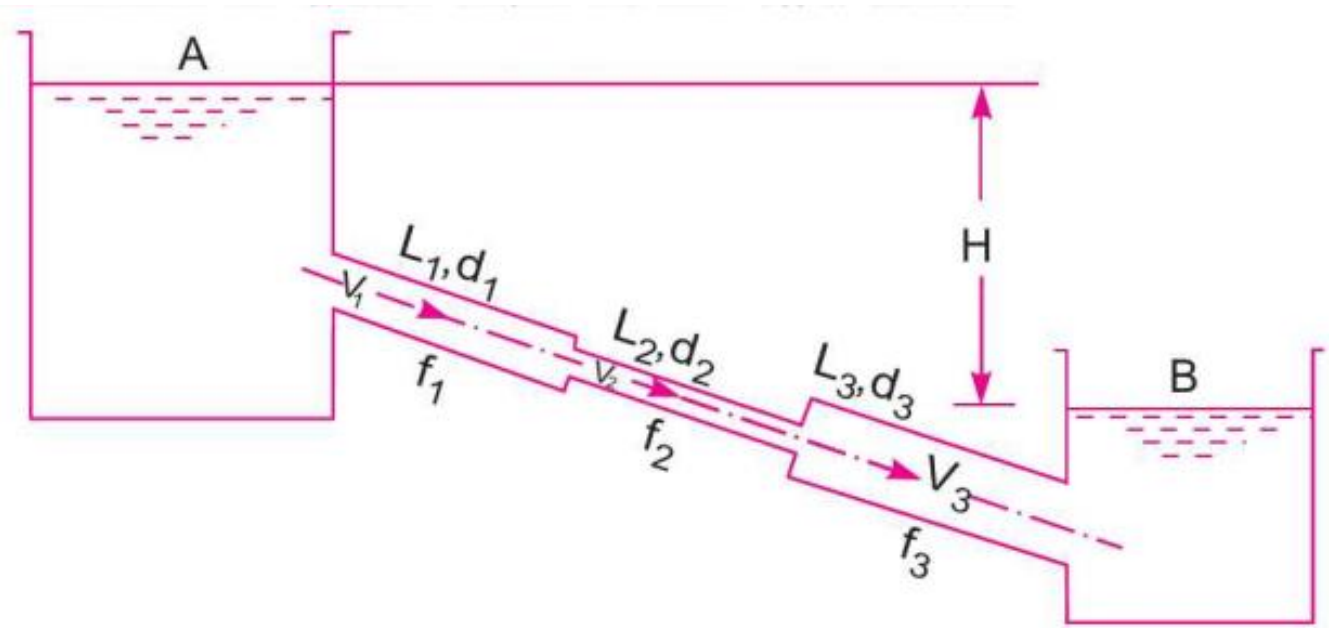
Equivalent pipe (for pipes in series)



$$V = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4}d^2} = \frac{4Q}{\pi d^2}$$

$$H = \frac{4f \,.\, L \,.\, V^2}{d \times 2g} \longrightarrow H = \frac{4\,f \,.\, L \,.\left(\frac{4Q}{\pi d^2}\right)^2}{d \times 2g} = \frac{4 \times 16 Q^2 f}{\pi^2 \times 2g}\left[\frac{L}{d^5}\right]$$

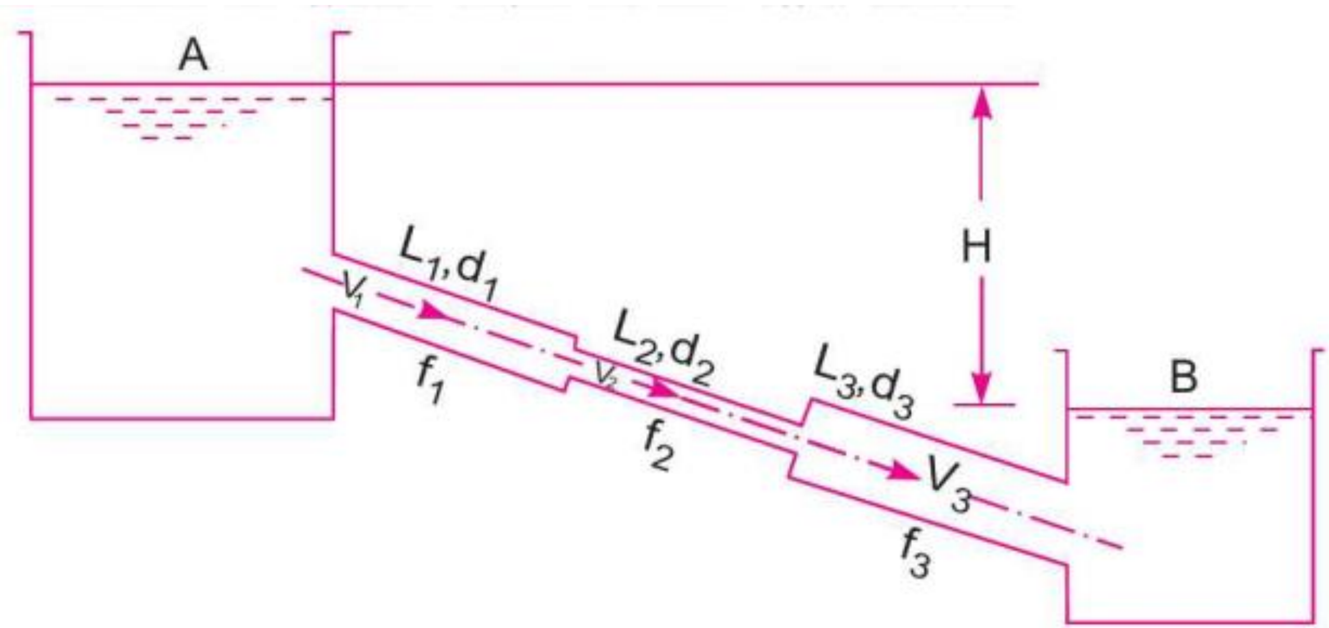
Equivalent pipe (for pipes in series)



$$H = \frac{4 \times 16 f Q^2}{\pi^2 \times 2g}\left[\frac{L_1}{d_1^5} + \frac{L_2}{d_2^5} + \frac{L_3}{d_3^5}\right]$$

$$H = \frac{4\, f \,.\, L \,.\left(\frac{4Q}{\pi d^2}\right)^2}{d \times 2g} = \frac{4 \times 16 Q^2 f}{\pi^2 \times 2g}\left[\frac{L}{d^5}\right]$$

Equivalent pipe (for pipes in series)



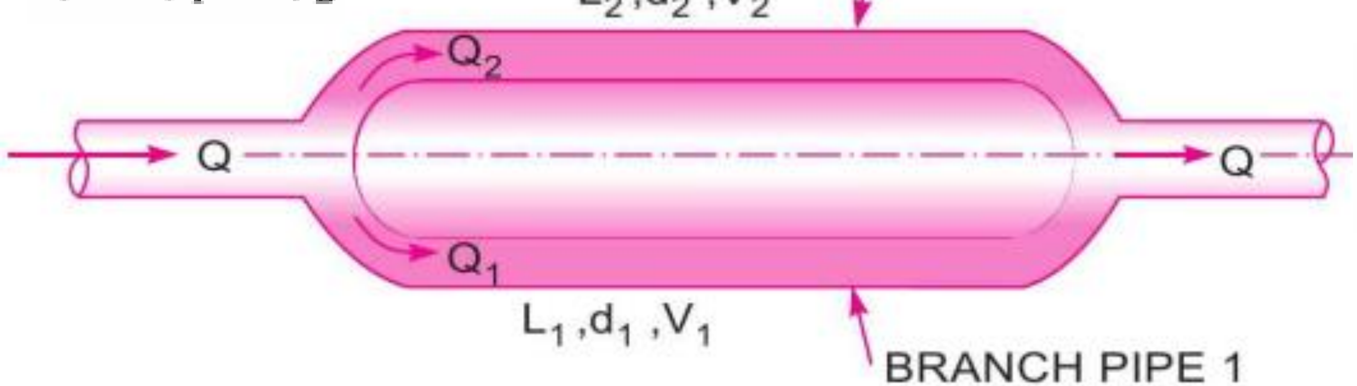
$$\frac{4\times 16 f Q^2}{\pi^2\times 2g}\left[\frac{L_1}{d_1^5}+\frac{L_2}{d_2^5}+\frac{L_3}{d_3^5}\right]=\frac{4\times 16 Q^2 f}{\pi^2\times 2g}\left[\frac{L}{d^5}\right]$$

$$\frac{L}{d^5}=\frac{L_1}{d_1^5}+\frac{L_2}{d_2^5}+\frac{L_3}{d_3^5}$$ (Dupuit's equation)

Flow through parallel pipes

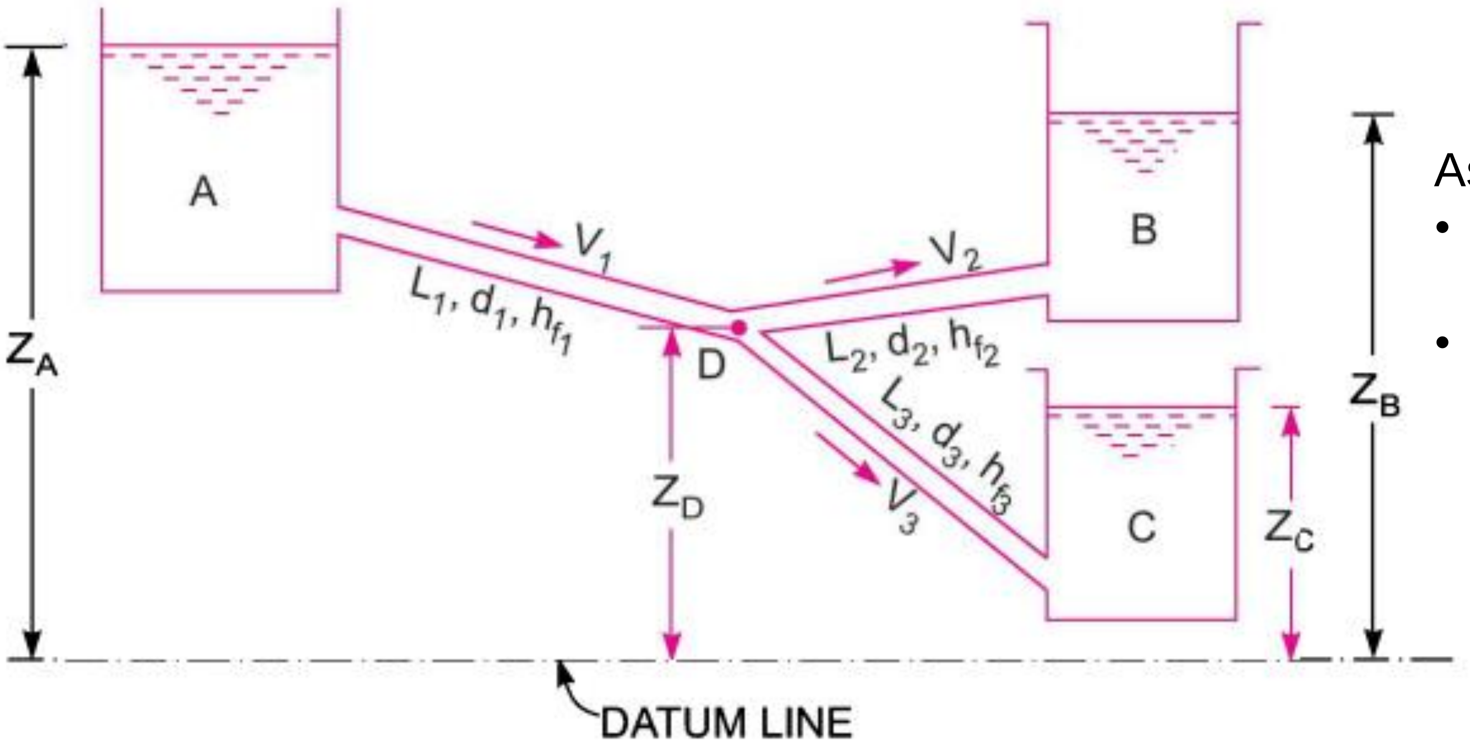
- Main pipe divides into two or more branches
- Discharge through main pipe = sum of discharges through branches
- Head loss in each branch remains same

$$Q = Q_1 + Q_2$$



$$\frac{4 f_1 L_1 V_1^2}{d_1 \times 2g} = \frac{4 f_2 L_2 V_2^2}{d_2 \times 2g}$$

Flow through branched pipes



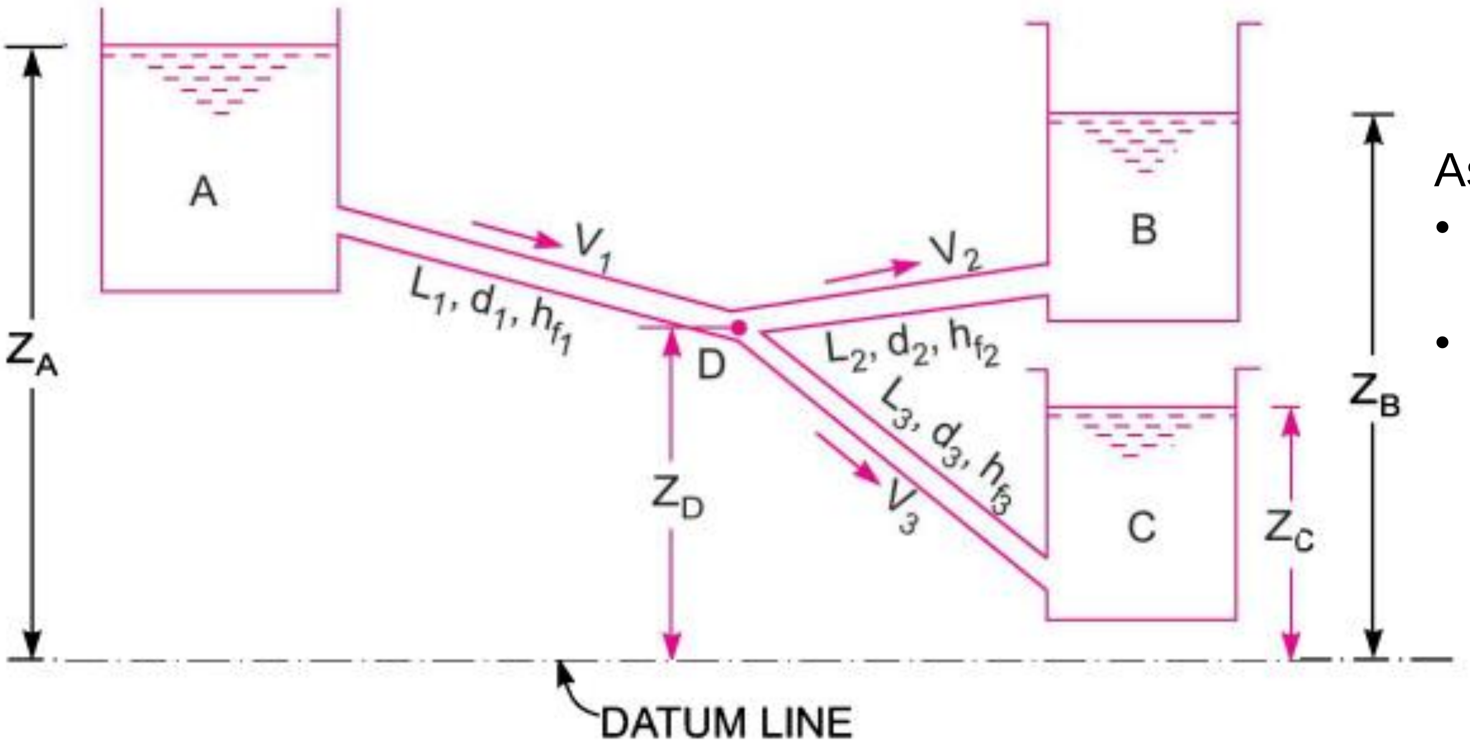
Assumptions:

- Reservoirs are large
- Minor losses are negligible as compared to major losses

Equations used to solve this problem

- Continuity → Inflow at junction = outflow at junction
- Bernoulli's equation
- Darcy-Weisbach equation

Flow through branched pipes



Assumptions:

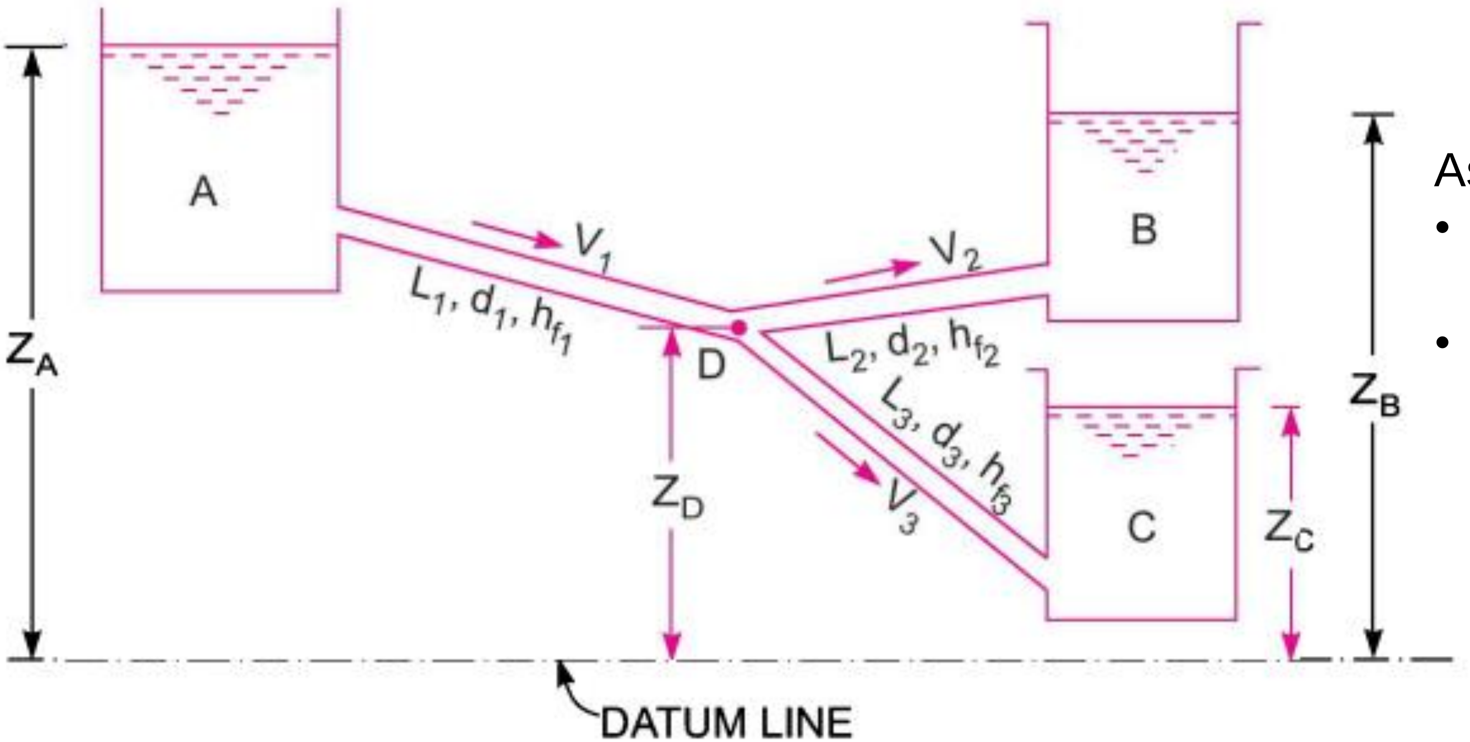
- Reservoirs are large
- Minor losses are negligible as compared to major losses

Equation #1

For flow from A to D from Bernoulli's equation

$$Z_A = Z_D + \frac{p_D}{\rho g} + h_{f_1}$$

Flow through branched pipes



Assumptions:

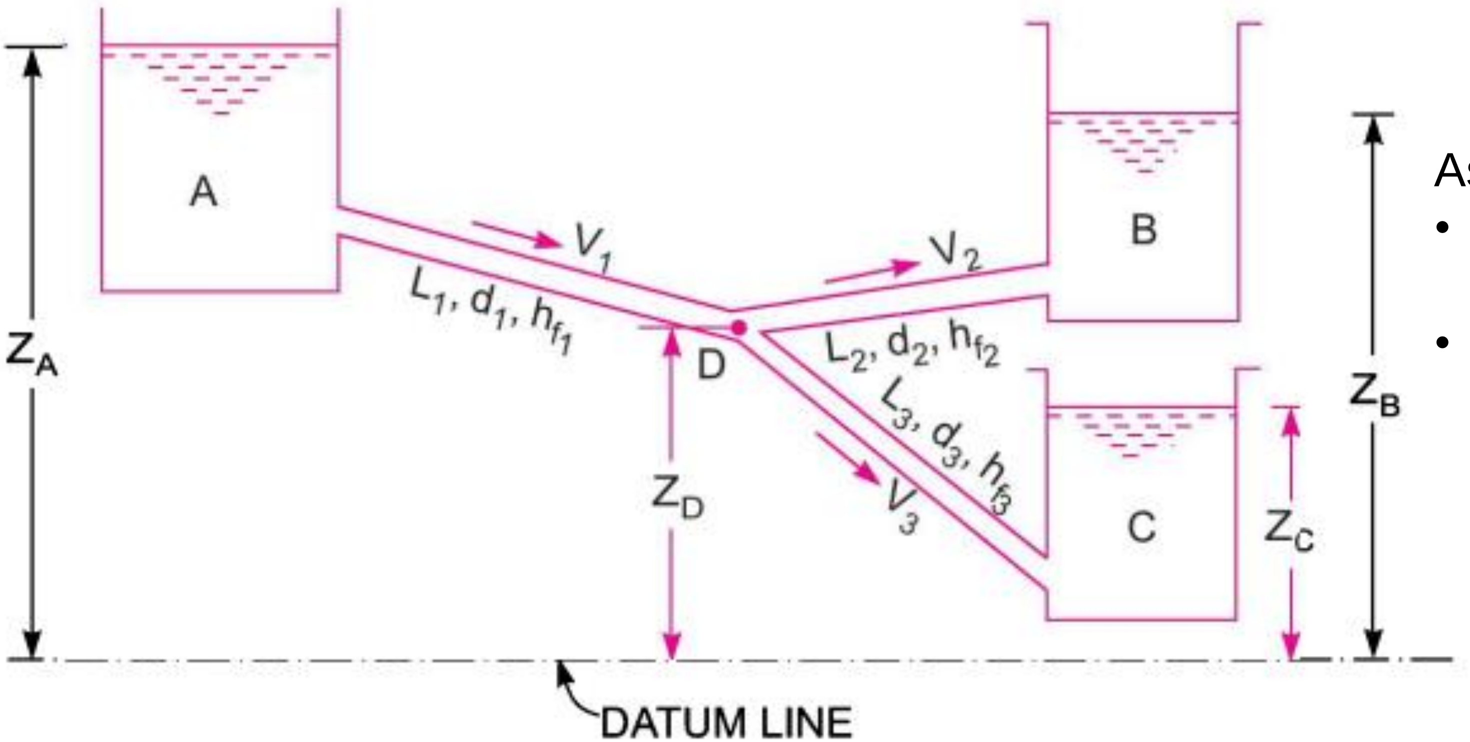
- Reservoirs are large
- Minor losses are negligible as compared to major losses

Equation #2

For flow from D to B from Bernoulli's equation

$$Z_D + \frac{p_D}{\rho g} = Z_B + h_{f_2}$$

Flow through branched pipes



Assumptions:

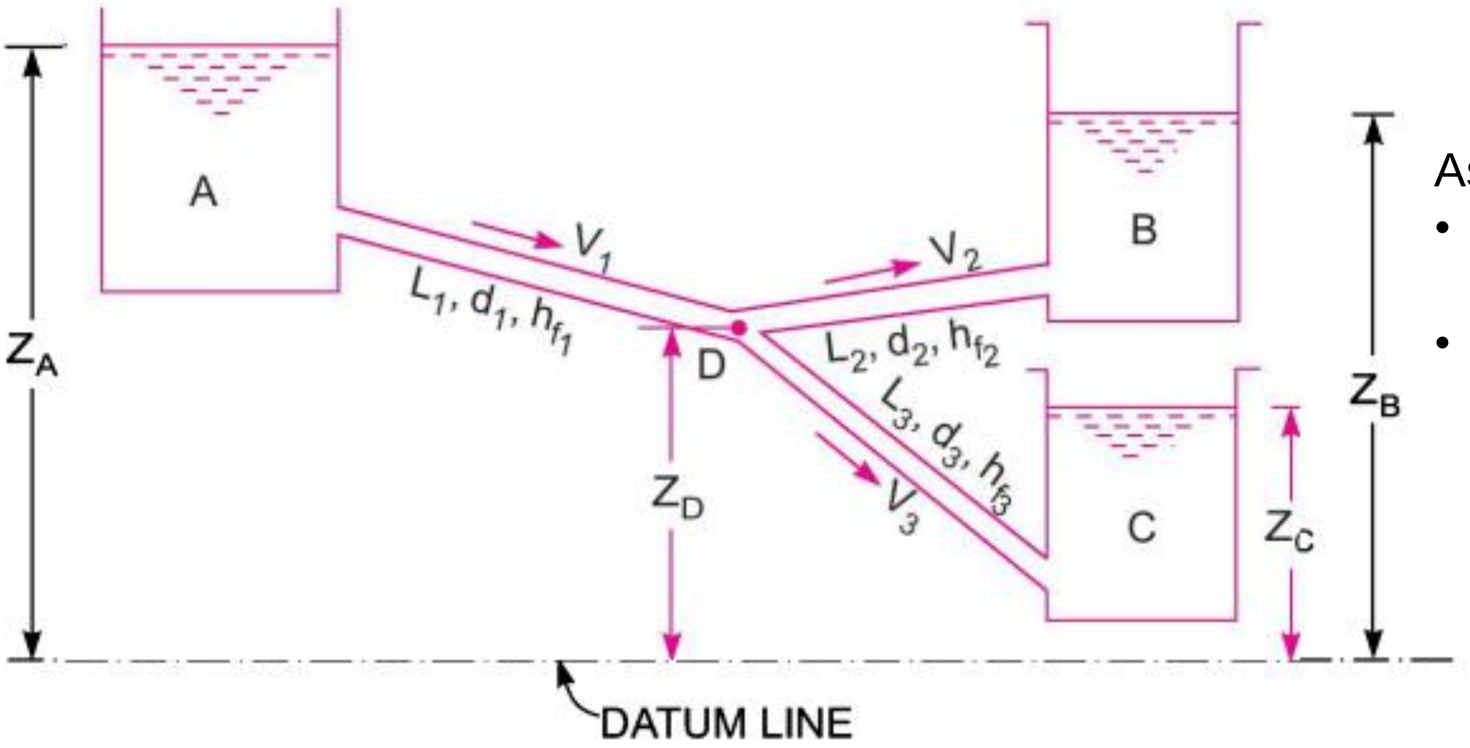
- Reservoirs are large
- Minor losses are negligible as compared to major losses

Equation #3

For flow from D to C from Bernoulli's equation

$$Z_D + \frac{p_D}{\rho g} = Z_C + h_{f_3}$$

Flow through branched pipes



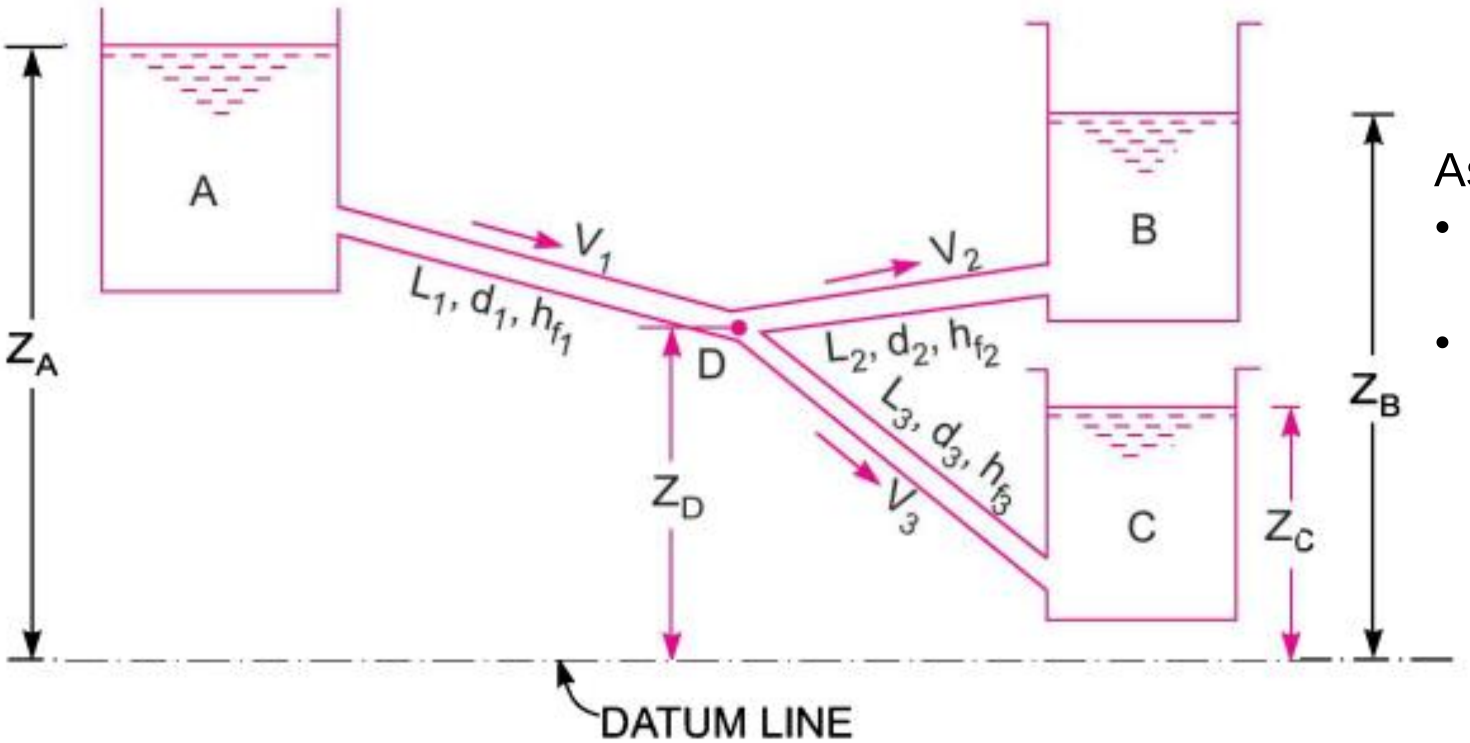
Assumptions:

- Reservoirs are large
- Minor losses are negligible as compared to major losses

Equation #4

$$\frac{\pi}{4} d_1{}^2 V_1 = \frac{\pi}{4} d_2{}^2 \times V_2 + \frac{\pi}{4} d_3{}^2 V_3$$

Flow through branched pipes



Assumptions:

- Reservoirs are large
- Minor losses are negligible as compared to major losses

There are four unknowns *i.e.*, V_1, V_2, V_3 and $\frac{p_D}{\rho g}$ and there are four equations

THANK YOU